EX NO: 2a

GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS USING MATLAB

AIM :

To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier according to the amplitude of the message signal, and to analyze the time-domain waveform and frequency spectrum of the FM signal using FFT in order to determine the carrier component, sidebands, frequency deviation, modulation index, and bandwidth based on Carson's Rule.

APPARATUS REQUIRED :

- MATLAB SOFTWARE

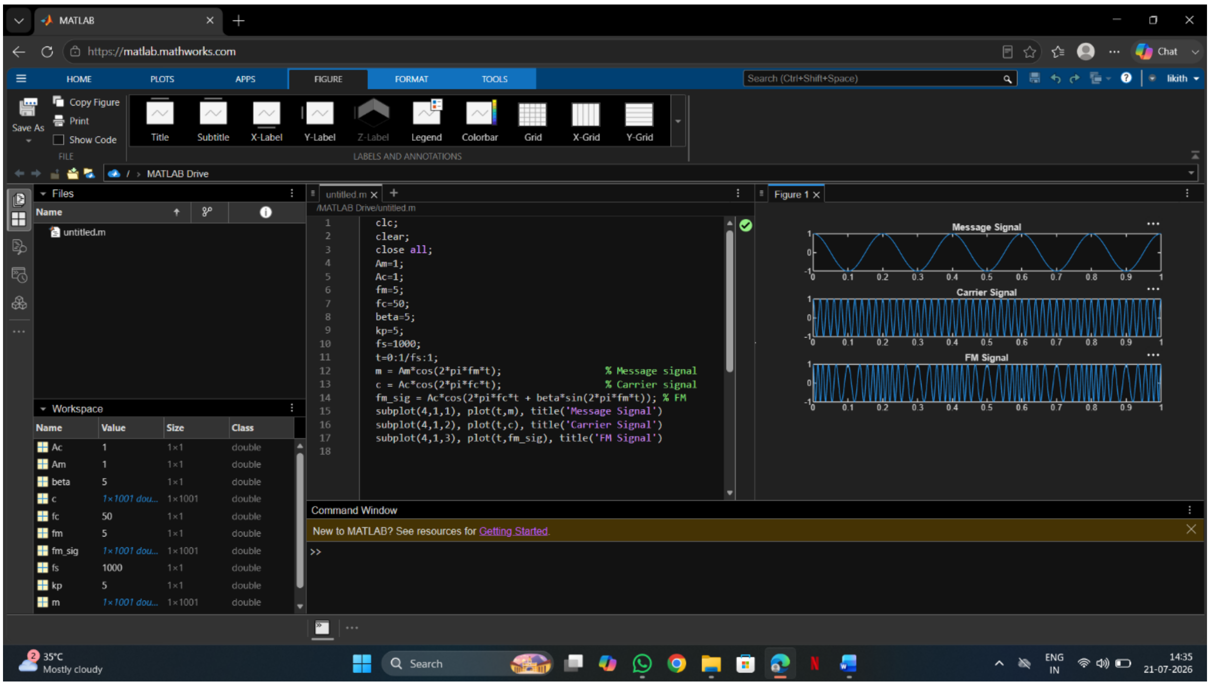
Objective 1:

To generate a Frequency Modulated (FM) signal using MATLAB by varying the frequency of a high-frequency carrier in accordance with the amplitude of the message signal and to observe the resulting waveform in the time domain.

PROGRAM (MATLAB CODE) :

```
clc;

clear;

close all;

Am   = 1;

Ac   = 1;

fm   = 5;

fc   = 50;

beta = 5;

 kp   = 5;

fs   = 1000;

 t = 0:1/fs:1;

m = Am * cos(2*pi*fm*t);      % Message signal

c = Ac * cos(2*pi*fc*t);           % Carrier signal

fm_sig = Ac * cos(2*pi*fc*t + beta*sin(2*pi*fm*t)); % FM

subplot(4,1,1), plot(t,m),      title('Message Signal')

subplot(4,1,2), plot(t,c),      title('Carrier Signal')

subplot(4,1,3), plot(t,fm_sig), title('FM Signal')
```

OUTPUT (GRAPH)



RESULT :

The Frequency Modulated (FM) signal was successfully generated using MATLAB, and the time-domain waveforms of the message signal, carrier signal, and FM signal were observed and analyzed.

Objective 2:

To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule

PROGRAM (MATLAB CODE)

% Calculation and Verification of Frequency Deviation

% and Modulation Index of FM Signals

clc;

clear;

close all;

%% Input Parameters

Am = 2;

% Message signal amplitude

fm = 500;

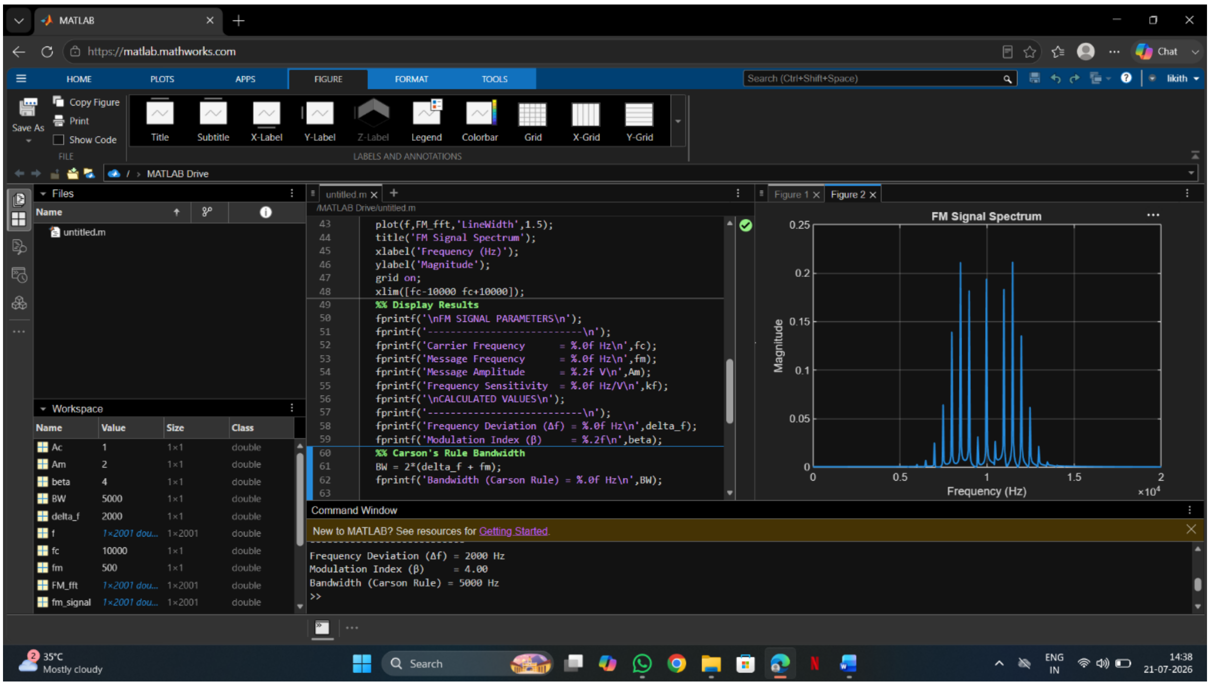
Ac = 1;

fc = 10000;

```
kf = 1000;
fs = 100000;
T  = 0.02;
t = 0:1/fs:T;
% Message frequency (Hz)
% Carrier amplitude
% Carrier frequency (Hz)
% Frequency sensitivity (Hz/V)
% Sampling frequency (Hz)
% Simulation duration (s)
%% Message Signal
m = Am * cos(2*pi*fm*t);
%% Frequency Deviation Calculation
delta_f = kf * Am;
%% Modulation Index Calculation
beta = delta_f / fm;
%% FM Signal Generation
fm_signal = Ac * cos(2*pi*fc*t + beta * sin(2*pi*fm*t));
%% Plot Message Signal
figure;
subplot(2,1,1);
plot(t, m, 'LineWidth', 1.5);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
%% Plot FM Signal
subplot(2,1,2);
plot(t, fm_signal, 'LineWidth', 1.5);
title('Frequency Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
```

```
grid on;
%% Spectrum Analysis
N      = length(fm_signal);
FM_fft = fftshift(abs(fft(fm_signal)) / N);
f      = (-N/2:N/2-1) * (fs/N);
figure;
plot(f, FM_fft, 'LineWidth', 1.5);
title('FM Signal Spectrum');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
xlim([fc-10000 fc+10000]);
%% Display Results
fprintf('\nFM SIGNAL PARAMETERS\n');
fprintf('---------------------------\n');
fprintf('Carrier Frequency      = %.0f Hz\n', fc);
fprintf('Message Frequency
fprintf('Message Amplitude
= %.0f Hz\n', fm);
= %.2f V\n', Am);
fprintf('Frequency Sensitivity  = %.0f Hz/V\n', kf);
fprintf('\nCALCULATED VALUES\n');
fprintf('---------------------------\n');
fprintf('Frequency Deviation (Δf) = %.0f Hz\n', delta_f);
fprintf('Modulation Index (β)     = %.2f\n', beta);
%% Carson's Rule Bandwidth
BW = 2 * (delta_f + fm);
fprintf('Bandwidth (Carson Rule) = %.0f Hz\n', BW);
```

OUTPUT :



RESULT :

i) The time-domain waveform was observed, and the frequency spectrum obtained using FFT has been plotted.

ii) Also, calculated the frequency deviation and bandwidth using Carson's rule.